

LAB 3 – ARCHITECTURE SELECTION & JUSTIFICATION

Inter-City Freight Load Matching Marketplace

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1. Architectural Choice

We chose Layered Architecture for the Inter-City Freight Load Matching Marketplace. The system is organized into Presentation, Business, and Data layers, providing a clear separation between the user interface, marketplace services, and stored data.

2. Reasons for the Choice

- Clear separation of responsibilities: The Presentation Layer handles interactions for shippers and freight carriers, the Business Layer handles load posting, bid management, and escrow payment processing, and the Data Layer stores marketplace information.
- Easy to understand and maintain: Separating marketplace functions into layers makes the system easier to organize, modify, and maintain as load matching, bidding, and payment requirements evolve.

3. Security Advantage

The layered structure separates the user-facing interface from the data storage layer. Access to sensitive marketplace and payment information can therefore be controlled through the Business Layer, reducing direct access to stored data and helping enforce security checks before data or payment operations are performed.

4. Performance Benefit

The separation of services allows frequently used operations such as load and carrier queries, bid processing, and payment-status updates to be handled by their respective business components. This avoids putting all processing into a single component and supports more efficient handling of marketplace requests.

Project Context

The marketplace is a B2B freight brokerage exchange where commercial shippers post cargo loads and trucking carriers submit bids. The system also supports automated milestone payment release after verified proof-of-delivery sign-off.